Data Handling: Import, Cleaning and Visualisation

Exercise/Workshop 1: Tools, Working with Text Files

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Prerequisites

- Nuvolos Account with RStudio

Exercises

Exercise A: Setting up a Working Environment

1. Get familiar with the file browser pane on the lower right. In Nuvolos, navigate to the folder "files". When you work on your PC locally, navigate to a location of your choice.
2. Use the 'New Folder'-button to create a new folder. Name this new folder `r_course`.
3. You should see the new folder listed in the file browser. Click on it to navigate to its contents. Now, click on the 'More' button and select 'Set as Working Directory' in the drop-down menu.
4. Again, use the 'New Folder'-button in order to create two new folders called `data` and `code`.
5. Finally, click on the project button in the top-right corner of the RStudio window and select 'New Project' in the drop-down menu. In the pop-up window, select 'Existing directory', browse to and select your `r_course` folder, then click 'Create Project'.

Now you know how to set up a meaningful basic folder structure and working environment for an R project. The next exercise teaches you how to write R scripts in this environment.

Exercise B: R Scripts

1. Switch to the R console and type the following line of code and hit enter (see example from above).

```r
print("Hello world")
```

You should see the words `"Hello world"` printed on screen. This is the usual way of working with R in an interactive session. However, in most circumstances, it makes sense to write the R code to an R script (in order to store and document it) and then execute the code from there.

2. In the RStudio Menu-bar select `File/New File/R Script` to create a new file, shown/opened in the Script pane.
3. Type `print("Hello world")` to the first line of the script, and click on 'Run' to execute the code in the console.
4. Save the file as `hello_world.R` (`File/Save As...`) in the sub folder `code` (created in the previous exercise).

5. Type the following command into the command line and hit enter:

```
source("code/hello_world.R")
```

```
source("../r_course/code/hello_world.R")
```

Now you know how to execute R code directly in the console (interactive session), how to execute lines of code written to an R script, as well as how to execute the entire R script (stored on disk) from the command line.

Exercise C: Saving output in a file

In Exercise B, you printed the phrase "Hello world" to your console. Now, you will save the phrase to a text file.

1. First, using the browser pane in RStudio, navigate to the sub folder `code` (created in Exercise A).
2. To make sure you are in the right folder, you can type `getwd()` into the console; it will show your current working directory. You should see `../r_course/code` as an output.
3. Now, let's save the output `"Hello world"` to a file named "hello_world_file.txt". Type the following command into the console:

```
cat("Hello world", file = "hello_world_file.txt")
```

4. In the file pane, open the folder `../r_course/code`. Make sure it contains a file named "hello_world_file.txt". You can open it to verify its content.

Important: throughout our exercise sessions, you will save output to .txt files and upload them to Github Classroom to complete the assignments. Make sure your .txt files are named according to our instructions - otherwise your assignment will fail.

Exercise D: Completing an assignment on GitHub classrom

1. In case you have not done so yet, create a GitHub account (for instructions, see the Notes to Exercise Session 1).
2. Once you have accepted the assignment (i.e., when you have clicked on the invitation link on Canvas), you will find yourself in your own repository for the current exercise session.
3. In your repository on GitHub, go to the `Add file` button and choose `Upload files`. Upload the "hello_world_file.txt" from the previous exercise and commit.
4. Next, navigate to the `Pull requests` item. If your submission was correct, you should see a green tick next to the `Feedback` icon, otherwise a red cross.
5. Click on `Feedback` to get more detailed info. Scroll down, click on `Merge pull request`, and confirm the merge.
6. Select the `Checks` item to compare your results to the solution.
7. In case your submission was not successful, you can create a new version of "hello_world_file.txt" and go through steps 3. to 6. again.

Important: throughout our exercise sessions, we will send you invitation links to assignments. They will always work in the same way (you upload an output file and can then request feedback automatically).

Additional Exercises (not exam-relavant)

Exercise E: Cloning and updating a GitHub repository

Remember, Git is a revision control system (to manage different versions of your code). GitHub is a hosting service for Git repositories. So Git is a tool, and GitHub a service for projects that use this tool.

- Clone the repository of this course. Cloning means that you create a local copy of the repo on your computer. Follow these instructions: https://help.github.com/en/articles/cloning-a-repository. (Alternatively, you could also download the entire repository as a zip-file using the graphical user interface, GUI, on the GitHub page.)